Exp.2 3/8/24

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

Aim:

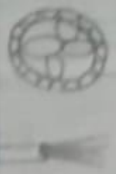
To study the different types of network cables

a) Understand different types of network cable

Different Type of cables used in networking are:

1. Unshielded Twisted Pair (UTTP) Cable
2. Shielded Twisted Pair (STP) cable
3. Co-axial Cable
4. Fibre optic cable

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / Use	Image
UTP	category 3	10bps	Advantages • Cheaper in cost • Easy to install as they have a smaller overall diameter disadvantages • More prone to (EMI) Electromagnetic interference and noise	10 Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, gigabit Ethernet	
	category 5	Upto 100 Mbps			
	category 5e	1Gbps			
STP	category 6, 6a	10 Gbps	Advantages • Shielded • Faster than UTP • Less susceptible to noise and	Gigabit Ethernet, 10 G Ethernet (55m) widely used in data centres	

			interference		
SSTP	Category 7	10 Gbps	Disadvantages • Expensive • Greater installation effort	Gigabit Ethernet 10 Gb Ethernet (100 m)	
Coaxial Cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages • High bandwidth • Immune to interference • Lower Bandwith • Versatile Disadvantages → Limited distance → cost → size is bulky	• Speed of signal is 80% of • Television Network • High speed internet connections	
Fibre optics cable	Single mode Multi mode	100Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantages • Expensive • Requires Skilled Installers	• Maximum distance of fibre optics cable is around 100 miles	

b) Make your own Ethernet Cross-Over Cable / Straight Cable

Tools and parts needed

- Ethernet cabling, CAT5e is certified for gigabit support, but CAT5 cabling work as well, just over shorter distances
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables
- Two RJ45 plugs
- Optional two plug shields

Step 1: To start construction of the device, begin by threading shields onto the cable

Step 2: Next strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round-d area to completly to this task

Step 3: After you will need to untangle the wires; there should be four "twisted pairs." Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other B.

Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown

Step 5: Next push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding. Simply strip the cables back a little more

Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cables) / using diagram A (to make a straightthrough cable)

To test it, plug it in and attempt to connect two devices directly

Student Observation:

1. What is the difference btwn cross cable and straight cable?
   Straight cables has the same wiring standard at both ends; cross cable has different wiring standards at each end
2. Which type of cable is used to connect two PC?
   To connect two PC cross cables are used
3. Which type cable is used to connect a router or switch to your PC?
   Straight cables
4. Find out the category of twisted pair cable used in your laptop to connect the PC to the network socket
   Usually cat5e is used
5. Write down your understanding, challenges faced and output recieved while making a twisted pair cable / straigh cable.

Understanding: Learned how to arrange and crimp the pair wires

Challenges: Maintaining the correct wire sequence and proper crimping

Output: Successfully made and tested the cable for network connectivity